

Medisignal Africa

AI-Powered Medicine Supply Chain Early-Warning and Intelligence Platform

Developer: Gabolekwe Topo Gabolekwe

Country: Botswana

Stage: Deployed and operational

Live demo: <https://medi-signal-africa.vercel.app>

Source code: <https://github.com/gabolekwetopo-bot/MediSignal-Africa>

The Problem

African health systems face medicine stockouts that are preventable but still kill patients every day. Existing Logistics Management Information Systems (LMIS) — OpenLMIS, mSupply, national platforms — report **what stock exists today**. They cannot answer **what will run out next**. Without prediction, ministries react to shortages instead of preventing them.

The Solution

Medisignal Africa is an **intelligence layer that sits above existing LMIS systems**. It ingests stock, consumption, and procurement data, then produces:

- **Shortage forecasts** up to 14 days ahead
- **Risk classification** per facility-medicine pair (Critical / High / Medium / Low)
- **Redistribution recommendations** matching surplus facilities to at-risk ones
- **Procurement risk flags** for orders arriving after projected stockout
- **Natural-language answers** via a grounded AI Supply Advisor
- **Printable situation reports** with AI-generated executive summaries

Medisignal does not replace LMIS. It makes them predictive.

How AI Is Used

AI Function	Purpose
Machine learning	Shortage forecasting, demand anomaly detection, risk scoring
Large language model	Conversational advisor grounded exclusively in live data

AI Function**Purpose**

Data analytics

Days-of-stock calculation, redistribution matching, hotspot detection

Responsible AI principles: human-in-the-loop for every action; no automated execution; "Insufficient Data" responses when facts are missing; full audit logging; confidence indicators on every prediction.

Current Deployment**Metric****Value**

Countries covered

4 (Botswana, Ghana, Kenya, Nigeria)

Health facilities

198

Essential medicines

55 (WHO essential list)

Consumption records

33,750 (6 months)

Procurement orders

210

Demo URL

<https://medi-signal-africa.vercel.app>**Key features demonstrated:**

- Interactive national shortage radar map
- AI Supply Advisor (streaming, grounded, country-aware)
- End-to-end procurement workflow: create → approve → dispatch → deliver
- AI-recommended redistribution with approval chains
- Data import with Excel templates and staging tables
- Role-based access design (6 roles)
- Alerts with one-click action
- 6-page PDF-ready situation report with AI executive summary

Current data: synthetic demonstration data while ministry pilot partnerships are established.

Alignment with SDGs and Agenda 2063

- **SDG 3** — Good Health and Well-Being
- **SDG 9** — Industry, Innovation and Infrastructure
- **SDG 10** — Reduced Inequalities
- **SDG 12** — Responsible Consumption and Production
- **SDG 17** — Partnerships for the Goals
- **Agenda 2063** — Goal 1 (Prosperous Africa) and Goal 3 (Healthy Citizens)

Scaling Plan (Next 24 Months)

1. Live LMIS integration agreements with national systems
2. Ministry pilot in at least one country (Botswana or Ghana priority)
3. Regulatory guidance for AI decision-support in public health
4. Expansion to 10–15 countries and 5,000+ facilities

Why Medisignal Should Be Selected

Medisignal is **not a concept** — it is a deployed, working platform. It is African-built, African-led, and designed specifically for African health systems. It addresses a problem that kills people every day. And it is already running in four countries with a fully functional AI advisor and government-grade reporting.

The ECA Innovation Café exists to showcase what African innovators are already doing. **Medisignal is already doing it.**

Contact

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